Lab Tasks

Task # 01:

You are working as a software engineer for a financial company. The company has a requirement to calculate the daily profit or loss of their stock portfolio. They have a list of the opening and closing prices of stocks for a given day.

- Write a Java program that takes two arrays of doubles as input, one containing the opening prices and the other containing the closing prices of the stocks, and returns an array of doubles representing the daily profit or loss for each stock in the portfolio.
- Your program should include a class named "PortfolioAnalyzer" with a function named "calculateDailyProfitLoss" that takes the input arrays of opening and closing prices as arguments and returns an array of doubles representing the daily profit or loss for each stock.
- For example, if the opening prices array is [10.0, 12.5, 8.0, 15.0] and the closing prices array is [11.0, 13.0, 7.0, 16.0], the program should return an array of doubles containing [1.0, 0.5, -1.0, 1.0], since the profit or loss for each stock can be calculated as the difference between the closing price and the opening price.

Note: You may assume that the input arrays have the same length, and that the order of the stocks in the opening prices array corresponds to the order of the stocks in the closing prices array.

Task # 02:

Write a Java program that takes an ArrayList of integers and removes all duplicates, leaving only the distinct elements in the list.

- Your program should include a class named "DuplicateRemover" with a static function named "removeDuplicates" that takes the input ArrayList as an argument and returns an ArrayList containing only the distinct elements.
- For example, if the input ArrayList is [1, 2, 3, 1, 4, 2, 5], the program should return an ArrayList containing [1, 2, 3, 4, 5].
- Note: You may not use any built-in Java functions or methods that directly remove duplicates from the input ArrayList. You should implement your own algorithm to solve this problem.

Task # 03:

Create an Array list that:

- Ask the user to enter 5 numbers as input

- Check if the list contains a prime number, then display that the list has prime numbers and display the sum of the numbers present at even indices like elements present at 0th, 2nd and 4th indices will be added.
- If the list doesn't contain prime numbers, then display the sum of the numbers present at odd indices like elements present at 1st and 3rd indices will be added.